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1 November 1961
SUPERSEDING
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10 April 1953

MILITARY SPECIFICATION

BATTERIES, PRIMARY, WATER-ACTIVATED (DUNK TYPE)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy and the Air Force.

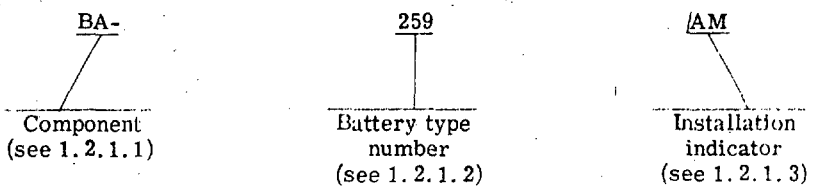
1. SCOPE

1.1 Scope. - This specification covers primary batteries which are inert until immersed in or filled with water for specified periods of activation and are discharged in air.

1.2 Classification. - Batteries shall be of the following types, as specified (see 6.1):

Type BA-253/U.
Type BA-259/AM.
Type BA-292/AM.
Type BA-316/AM.

1.2.1 Type designation. - The type designation of water-activated primary batteries shall be in the following form:



1.2.1.1 Component. - Primary batteries are identified by two-letter symbol "BA" followed by a hyphen.

1.2.1.2 Battery type number. - The battery type number identifies the specific battery.

1.2.1.3 Installation indicator. - The installation indicator is composed of one or more letters preceded by a slant line, denoting the equipment with which the battery is associated, as specified herein.

/AM - Airborne meteorological equipment.
/U - General utility (includes) two or more installation classes: airborne, shipboard, and ground.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-B-626 - Brass, Leaded and Nonleaded; Rods, Shapes, Forgings and Flat Products With Finished Edges (Bars, Flat Wire and Strips).
QQ-C-576 - Copper Plates, Rolled Bars, Sheets, and Strips.
QQ-P-330 - Phosphor Bronze Bars, Plates, Rods, Sheets, Strips, Flat Wire, and Structural and Special Shaped Sections.
UU-C-201 - Cardboard, Bristol, Manila, and Railroad.
PPP-B-576 - Box, Wood, Cleated, Veneer Paper Overlaid.

FSC 6135

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FEDERAL (cont'd.)
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiber.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-C-843 - Cushioning Material, Cellulosic.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

MILITARY
MIL-W-76 - Wire and Cable, Hook-Up, Electrical, Insulated.
MIL-P-116 - Preservation, Methods of.
MIL-T-713 - Twine and Tape, Lacing and Tying (for Use in Electrical and Electronic Equipment).
MIL-P-3115 - Plastic-Material, Laminated, Thermosetting, Sheets, Paper-Base, Phenolic Resin.
MIL-S-6872 - Soldering Process, General Specification for.
MIL-L-10547 - Liners, Case, Waterproof.

STANDARDS

FEDERAL
FED-STD-595 - Colors.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.

CONSOLIDATED CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 1 Park Avenue at 33rd St., New York 16, N. Y.)

3. REQUIREMENTS

3.1 Preproduction inspection. - Prior to production, preproduction samples as specified (see 6.1) shall be forwarded to a laboratory satisfactory to the bureau or agency concerned for the tests specified in 4.5.

3.2 Material. - The material for each part shall be as specified herein. When a definite material is not specified, a suitable material shall be used. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished battery.

3.2.1 Wire. - When insulated wire is used for electrical connectors and terminal leads, it shall conform to Specification MIL-W-76.

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3.2.2 Brass. - Brass used for purposes other than internal construction of the cell shall conform to Specification QQ-B-626.

3.2.3 Copper. - Copper used for purposes other than internal construction of the cell shall conform to Specification QQ-C-576.

3.2.4 Phosphor bronze. - Phosphor bronze used for purposes other than internal construction of the cell shall conform to Specification QQ-P-330.

3.3 Design and construction. - The batteries shall be of the design, construction, physical dimensions, weight, voltages, and polarity specified (see 3.7). The battery shall be constructed for simple and rapid activation for service. The total time for activating and preparing the battery for use shall not exceed the specified time (see 3.7).

3.3.1 Internal connections. - All connections shall be securely made. Connections between cells may be made with bare wire or suitable equivalent. Wire soldered to terminal lugs shall not be bared more than one-eighth of an inch from the lugs, nor extended more than one-eighth inch beyond the lugs.

3.3.2 Terminals. - The type, dimensions, and spacing of terminals shall be as specified (see 3.7). Terminals shall provide a positive means of making and maintaining external electrical connections to the battery. Contact surfaces of the terminals shall not be obstructed by wax or other insulating compounds.

3.3.2.1 Socket-type terminals. - Contact portions of socket-type terminals shall be made of spring phosphor bronze or plated spring brass. Sockets shall be of such design, construction, and material as to minimize leakage, current between terminals and to prevent the short circuiting of terminals, or even momentary contact of any of the pins of the mating plugs to any socket terminal other than those for which such pins are intended. The socket shall provide a firm and positive contact with the pins of the mating plug. Unless otherwise specified (see 3.7), the pin-hole-spacing tolerances for socket terminals shall be plus or minus 0.005 inch. When the socket is not mounted directly on the battery jacket, it shall be connected with cable (hook-up wire) conforming to type MW of Specification MIL-W-76. The wires leading from the battery to the socket shall be twisted or bound together to prevent shifting or unraveling, and the ends of the lacing twine, if used, shall be tied to prevent loosening. When lacing twine is used for binding the wire together, it shall conform to type N, class 3 of Specification MIL-T-713.

3.3.2.2 Wire-lead terminals. - Unless otherwise specified (see 3.7), wire-lead terminals shall be of cable (hook-up wire) conforming to type MW of Specification MIL-W-76. The free end of each wire shall be bared for a distance of 1/2 ± 1/8 inch. The strands of each wire so bared shall be twisted together and soldered.

3.3.2.3 Terminal markings. - Terminal markings shall be as specified (see 3.7). Polarity, voltage, and unit of the battery (A, B, C) shall be legibly marked.

3.3.3 Jackets. - The jacket is defined as the outer container housing the battery contents. The jacket shall be waterproof and fabricated in such a manner as to prevent its coming apart during activation or when subjected to any of the specified tests (see 4.7.5).

3.3.3.1 Color of jackets. - Unless otherwise specified (see 3.7), the color of all exposed surfaces of jackets shall conform to lusterless green No. 34087 of Standard FED-STD-595.

3.3.4 Threaded parts. - All threaded parts shall be in accordance with Handbook H28.

3.4 Capacity and voltage. - When tested in accordance with 4.7.5, voltage of the battery, or any unit of the battery (in batteries composed of more than one unit), shall not exceed the maximum specified voltage nor fall below the specified test-end voltage within the time (capacity) specified. If limits of voltage fluctuations are specified, they shall not be exceeded (see 3.7).

3.5 Labeling and marking. - All labeling and marking shall remain clear and legible throughout all of the specified tests. The color of lettering shall be black.

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3.5.1 Identification label. - Each battery shall have an identification label positioned as specified (see 3.7). Unless otherwise specified (see 3.7), there shall be no information on the label other than the following:

Type designation
Order number
Code number
Manufacturer's name
(Trade-mark may also be used)

Example:

Battery BA-253/U
Order number
8187-PHILA-51-15-16
0751
John Doe Company
JODOC

3.5.1.1 Code number. - The code number shown shall indicate the month and year of manufacture of the battery by means of a four-digit number in which the first two digits shall indicate the month and the last two digits the year. Months earlier than the tenth month shall be preceded by "O". When a battery is completed during the last ten working days of a month or the first ten working days of the subsequent month, the manufacturer is permitted to use either month as the coded month of manufacture.

3.5.1.2 Instruction label. - Each battery shall be provided with an instruction label, positioned as specified (see 3.7). The instruction label shall contain complete instructions for activating and preparing the battery for service. The instructions shall be approved by the bureau or agency concerned.

3.6 Workmanship. - The batteries shall be manufactured and processed in a careful and workmanlike manner in accordance with good design and sound construction.

3.6.1 Soldering. - Soldering operations shall be performed in accordance with Specification MIL-S-6872.

3.7 Detail requirements. -

3.7.1 Type BA-253/U. -

3.7.1.1 Electrical and physical requirements. - The electrical and physical requirements shall be as follows:

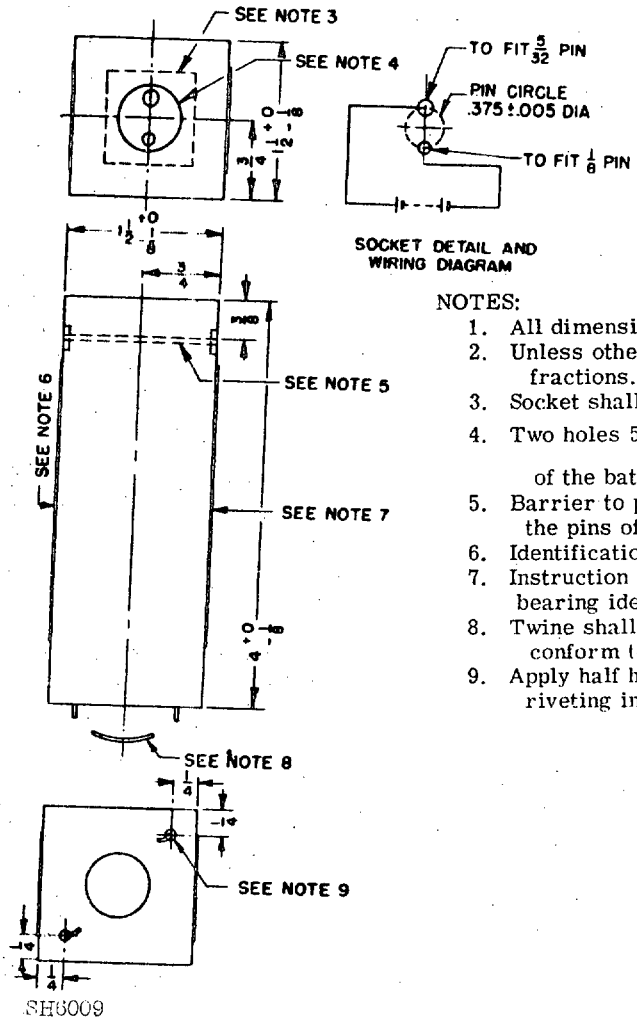
Nominal voltage	6.0 volts
Maximum voltage during discharge:	
At high and low temperature	6.8 volts
At normal temperature	7.9 volts
Test-end voltage	5.5 volts
Time for activation (maximum)	15 minutes
Resistance load	12 ohms
Terminals	Socket type
Weight (activated)	80 ± 10 grams
Drop test	When tested as specified herein, battery suspension cord shall not rip or separate from its anchorage, and battery shall not experience any reduction in capacity.
Capacity and voltage tests (minimum time to reach test end voltage):	
Initial low-temperature discharge	45 minutes
Initial high and low temperature discharge	45 minutes
Initial normal temperature discharge	32 minutes

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Capacity and voltage tests (minimum time to reach test end voltage): (Cont'd.)

Delayed low temperature discharge	45 minutes
Delayed high and low temperature discharge	45 minutes
Delayed normal temperature discharge	32 minutes

3.7.1.2 Socket detail and wiring diagram. - The socket detail and wiring diagram shall be as shown on figure 1.



NOTES:

1. All dimensions in inches.
2. Unless otherwise specified, tolerances are ±1/32 on fractions.
3. Socket shall be No. 2636 Cinch Mfg. Corp., or equal.
4. Two holes 5/8 $^{+1/16}_{-1/32}$ diameter, one centered on each end of the battery jacket.
5. Barrier to prevent activating water from short-circuiting the pins of socket.
6. Identification label, centered on side of battery.
7. Instruction label, centered on opposite side of battery bearing identification label.
8. Twine shall be 2-1/2 ± 1/2 long between knots and shall conform to type N, class 3 of Specification MIL-T-713.
9. Apply half hitch and pull both ends of twine tight before riveting in place.

Figure 1 - Socket detail and wiring diagram for type BA-253/U.

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3.7.2 Type BA-259/AM. -

3.7.2.1 Electrical and physical requirements. - The electrical and physical requirements shall be as follows:

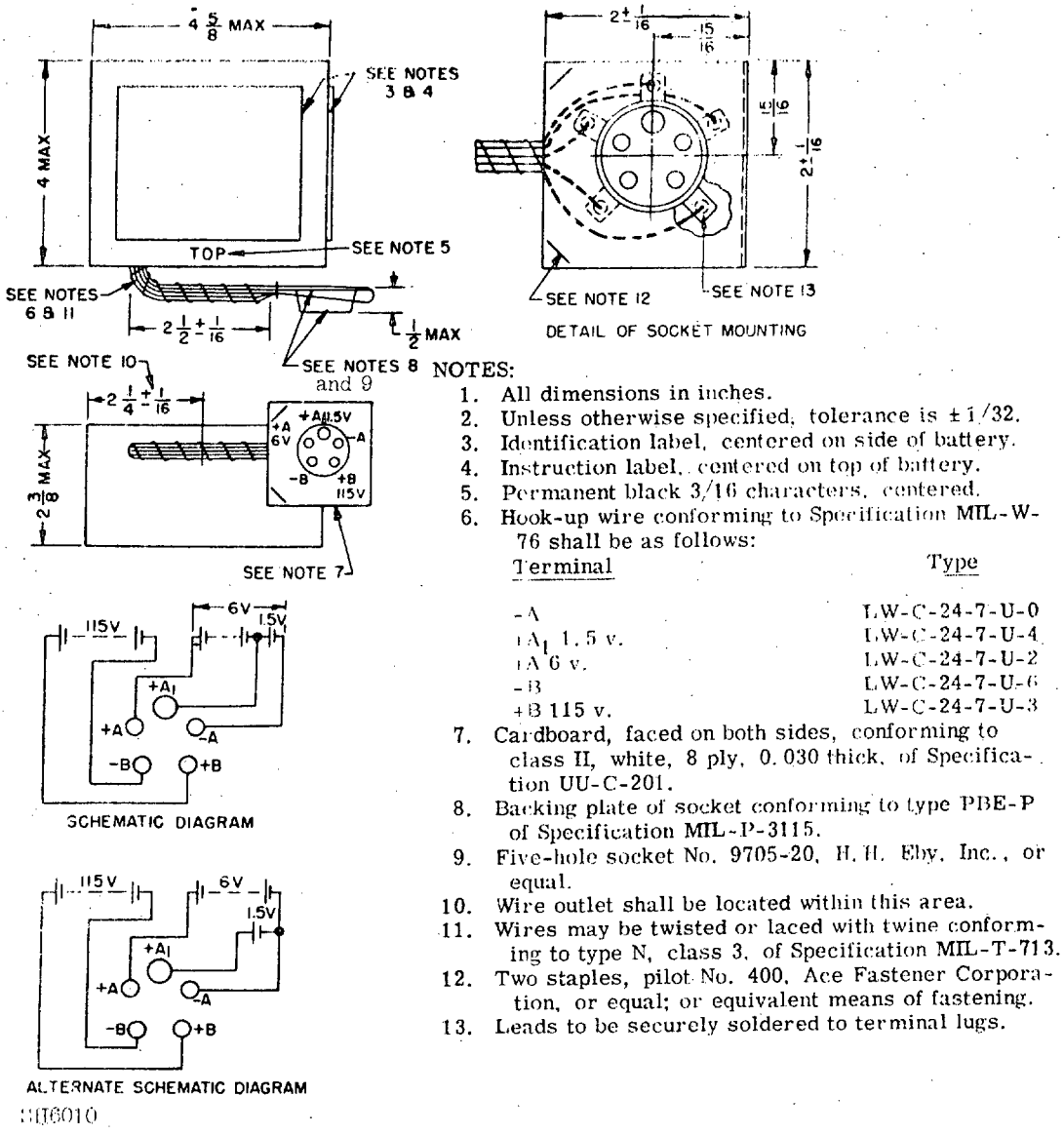
Unit	Nominal	Voltage of unit				Fluctuation 1/
		Maximum during discharge at		Minimum		
		High and low temperature	Normal temperature	After activation	Test-end	
A_1	1.5	1.5	1.7	1.1	1.1	± 0.01
A	6	7.4	7.9	5.5	5.5	± .05
B	115	130.0	130.0	95.0	95.0	± 1.00

Time for activation:	
On open circuit	10 minutes
On closed circuit	15 minutes
Total activation, maximum	25 minutes
Resistance load:	
A_1 unit	10.00 ± 0.06 ohms
A unit	28.80 ± 0.14 ohms
B unit	3050 ± 15 ohms
Terminals	Socket
Weight (activated), maximum	450 grams
Capacity and voltage tests (minimum time to reach test end voltage)	
Initial low-temperature discharge	130 minutes
Initial high and low temperature discharge	130 minutes
Initial normal temperature discharge	90 minutes
Delayed low-temperature discharge	130 minutes
Delayed high and low temperature discharge	130 minutes
Delayed normal temperature discharge	90 minutes

1/ During any of the discharge tests.

3.7.2.2 Socket detail and wiring diagram. - The socket detail and wiring diagram shall be as shown on figure 2.

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NOTES:

1. All dimensions in inches.
2. Unless otherwise specified, tolerance is ±1/32.
3. Identification label, centered on side of battery.
4. Instruction label, centered on top of battery.
5. Permanent black 3/16 characters, centered.
6. Hook-up wire conforming to Specification MIL-W-76 shall be as follows:

Terminal	Type
-A	LW-C-24-7-U-0
+A1 1.5 v.	LW-C-24-7-U-4
+A 6 v.	LW-C-24-7-U-2
-B	LW-C-24-7-U-6
+B 115 v.	LW-C-24-7-U-3

7. Cardboard, faced on both sides, conforming to class II, white, 8 ply, 0.030 thick, of Specification UU-C-201.
8. Backing plate of socket conforming to type PBE-P of Specification MIL-P-3115.
9. Five-hole socket No. 9705-20, H. H. Eby, Inc., or equal.
10. Wire outlet shall be located within this area.
11. Wires may be twisted or laced with twine conforming to type N, class 3, of Specification MIL-T-713.
12. Two staples, pilot No. 400, Ace Fastener Corporation, or equal; or equivalent means of fastening.
13. Leads to be securely soldered to terminal lugs.

Figure 2 - Socket detail and wiring diagram for type BA-259/AM.

3.7.3 Type BA-292/AM. -

3.7.3.1 Electrical and physical requirements. - The electrical and physical requirements shall be as follows:

Nominal voltage	3.2 volts
Maximum voltage during discharge	3.8 volts

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Test-end voltage	3.0 volts
Time for activation (maximum)	1.0 minutes
Resistance load	1 lamp[1/]
Terminals	Wire leads (attached to lamp)
Weight (activated)	25 ± 5 grams
Drop test	When tested as specified herein, battery suspension cord shall not rip or separate from its anchorage, and battery shall not experience any reduction in capacity.
Capacity and voltage tests (minimum time to reach test and voltage)	
Initial high and low temperature discharge	38 minutes
Delayed high and low temperature discharge	38 minutes

1/ Lamp shall be number 366, Westinghouse Electric Corp., or equal, attached to the battery by soldering leads directly to lamp or other means providing secure and positive contact throughout discharge, while the battery is in an inverted vertical position.

3.7.3.2 Overall dimensions. - The overall dimensions for the battery shall be as follows:

	Inches (maximum)
Length	1-1/2
Width	1-1/2
Height	3

The configuration of the battery is optional.

3.7.3.3 External thermal insulation, when required, shall be wood fiber, cotton, or any other material having similar thermal properties with the exception of fiber glass or glass wool.

3.7.3.4 A tab of plastic, fiber, or metal material of a minimum thickness of 1/32 inch having a 3/32 ± 1/32 inch hole in it shall be securely fastened to the bottom of the battery. Minimum distance from the edges of the tab to the circumference of the hole shall be 1/32 inch for metal material and 3/32 inch for plastic and fiber material.

3.7.3.5 Due to limited space on the battery, the instruction label shall appear on the unit-package container.

3.7.4 Type BA-316/AM. -

3.7.4.1 Electrical and physical requirements. - The electrical and physical requirements shall be as follows:

Unit	Voltage of unit				
	Nominal	Maximum during discharge	Minimum		Maximum fluctuation[1/]
			Initial	Test-end	
A	7.0	7.5	5.5	5.5	0.07
B	120	120	100	100	3.6

See footnote at end of table.

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Unit	Voltage of unit				
	Nominal	Maximum during discharge	Minimum		Maximum fluctuation 1/
			Initial	Test-end	

Time for activation, maximum	20 minutes
Resistance load:	
A unit	19 ohms
B unit	30,000 ohms
Terminals	Socket type
Weight (activated) maximum	450 grams
Capacity and voltage tests (minimum time to reach test end voltage)	
Initial normal temperature discharge	90 minutes
Initial low-temperature discharge	100 minutes
Initial flight similitude discharge	100 minutes
Delayed normal temperature discharge	90 minutes
Delayed low temperature discharge	100 minutes
Delayed flight-similitude discharge	100 minutes

1/ During any of the discharge tests (fluctuations are defined as repetitive voltage changes at intervals of less than 5 second duration).

3.7.4.2 Socket detail and wiring diagram. - The socket detail and wiring diagram shall be as shown on figure 3.

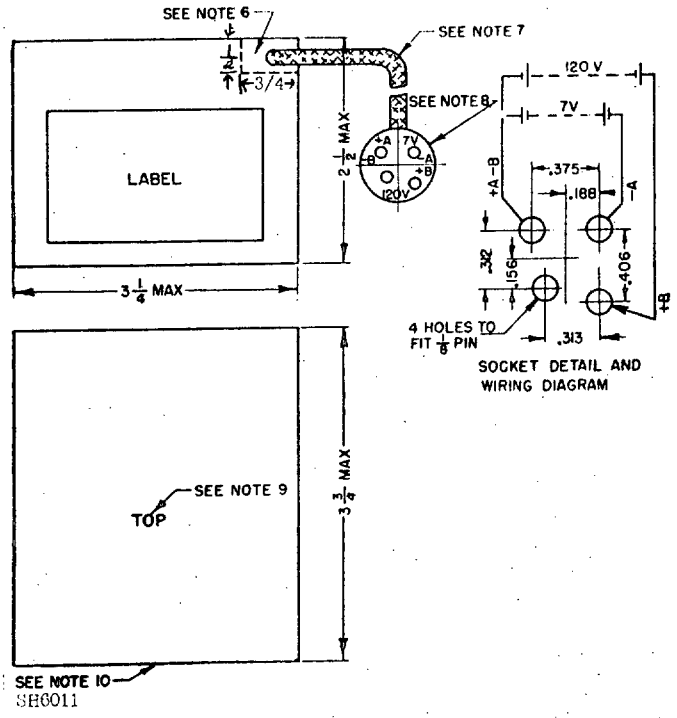


Figure 3 - Socket detail and wiring diagram for type BA-316/AM.

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Notes to figure 3:

1. All dimensions in inches.
2. All tolerances are ± 0.005 on decimals.
3. Each battery shall be accompanied by an instruction sheet of activating instructions in lieu of an instruction label. For preproduction inspection, six copies of instruction sheets shall accompany the preproduction inspection samples.
4. A plastic bag made of polyethylene or other similar waterproof plastic material and having a uniform thickness of 4.0 ± 0.5 mils shall be provided with each battery. This bag shall be capable of having the battery placed in it and its open end shall extend a minimum of 4 inches beyond the battery when the battery is in the bag.
5. The battery shall meet all of the specified electrical and capacity and voltage requirements when discharged in either of the following positions: When a panel marked "END" faces downward or when panel marked "TOP" faces upward.
6. Wire outlet shall be located within this area.
7. Lead wires from battery to socket shall be 12 ± 1 long. The lead wires shall conform to type HW-C20(7)U of Specification MIL-W-76.
8. Sockets shall be No. 2874, Cinch Mfg. Corp., or equal, to fit plug No. M64, Cinch Mfg. Corp., or equal. Polarity, voltage, and unit (A or B) of each socket terminal shall be clearly marked on the socket. If the socket is of the socket shield and receptacle type, this information may be shown in a drawing on the instruction sheet rather than on the socket itself. The lead wires shall be soldered to the appropriate socket terminals, and each soldered joint shall be insulated with an insulating paint or varnish so that no current leakage paths are possible when the battery is activated.
9. The panel marked "TOP" shall be easily opened for activation, and shall be easily replaced and secured after activation.
10. The word "END" shall be marked on this panel.

3.8 Minimum performance index (MPI). - The MPI of a battery shall be computed according to the following formula:

$$MPI = (DL + DHL + DN + DFS)$$

Where:

DL = Specified minimum number of minutes on delayed low-temperature discharge (when test is not specified DL = 0)

DHL = Specified minimum number of minutes on delayed high-and low-temperature discharge (when test is not specified DHL = 0)

DN = Specified minimum number of minutes on delayed normal temperature discharge (when test is not specified DN = 0)

DFS = Specified minimum number of minutes on delayed flight similitude discharge (when test is not specified DFS = 0)

Example: To calculate the MPI of battery BA-259/AM:

In 3.7.2.1 the specified minimum discharge time requirements are:
DL = 130 minutes
DHL = 130 minutes
DN = 90 minutes

Substituting test values in the above formulas, the following is obtained:

MPI = 130 + 130 + 90 = 350

3.8.1 Actual performance index (API). - The API of a battery shall be computed by substituting in formula (see 3.8) the average number of minutes on discharge obtained from all the sample batteries allocated and subjected to each capacity-voltage test under group C inspection (see 4.7.5) or the specified capacity value, whichever is lowest. The API shall be computed for each release.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - The examination and testing of batteries shall be classified as follows:

(a) Preproduction inspection (see 4.5 and 4.7).
(b) Acceptance inspection (see 4.6 and 4.7).

4.3 Standard test and storage conditions. -

4.3.1 Test conditions. - Unless otherwise specified (see 3.7), all tests shall be conducted at a temperature of 75 ± 10°F. (23.9 ± 5.6°C.).

4.3.2 Storage conditions. - All batteries intended for capacity and voltage tests shall be stored in their shipping containers at the temperature and relative humidity specified (see 3.7), within 2 weeks after date of receipt at the designated laboratory.

4.4 Test equipment and inspection facilities. - Test equipment and inspection facilities shall be of sufficient accuracy and quality to permit performance of the required inspection. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

4.4.1 Instrument accuracy. -

4.4.1.1 Voltmeters. - All batteries subjected to a test load shall be read with a voltmeter having a resistance of at least 1,000 ohms per volt. All voltmeters used shall be accurate within 1 percent of the full-scale reading. The voltmeter range shall be such that the readings are taken on the upper half of the scale.

4.4.1.2 Resistor tolerances. - In all tests involving discharge through a resistance, such resistance shall be of the following accuracy:

(a) Resistances from zero to 25,000 ohms shall be accurate within 0.5 percent.
(b) Resistances from above 25,000 ohms to and including 1 megohm shall be accurate within 1 percent.
(c) Resistances above 1 megohm shall be accurate within 5 percent.
(d) In determining the resistance used as a test load, the resistances of all continuously operating voltmeters used shall be considered as part of the specified resistance load (see 3.7).

4.4.1.3 Timing. - Timing equipment shall be accurate within 0.5 percent.

4.5 Preproduction inspection. - Preproduction inspection shall consist of the tests specified in tables II, III, IV and V for test groups I and II.

4.6 Acceptance inspection. -

4.6.1 Lot. - All batteries of the same type designation offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.6.2 Group A inspection. - Group A inspection shall consist of visual and dimensional examination on sample batteries.

4.6.2.1 Sampling for group A inspection. - Sample batteries shall be selected at random from each lot in accordance with Standard MIL-STD-105 at Inspection Level II. The acceptable quality level shall be 1.5 percent defective.

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4.6.3 Group B inspection. - Group B inspection shall consist of capacity and voltage tests under low-temperature discharge conditions (except BA-292/AM employing high and low temperature discharge conditions) without a previous storage period, and the selection and shipment of sample batteries to a laboratory satisfactory to the bureau or agency concerned with group C inspection.

4.6.3.1 Sampling for group B inspection. - Sample batteries shall be selected at random from the quantity on release. The quantity on release is the quantity of batteries of any one type dated as of any one month on any one contract or order. Sampling shall be in accordance with section B, inspection level III of Standard MIL-STD-414.

4.6.4 Group C inspection. - For group C inspection, the number of batteries selected from an acceptable lot shall be as specified in table I. Packaging shall be in accordance with 5.1, and all packages shall be sealed as specified therein. The manufacturer shall supply the sample batteries in addition to the number called for in the contract or order at no extra cost to the Government.

Table I - Sample batteries for group C inspection.

Quantity on release (see 4.6.1)	Number of sample batteries	Quantity on release (see 4.6.1)	Number of sample batteries
1 to 200	0	30,001 to 35,000	56
201 to 5,000	26	35,001 to 40,000	58
5,001 to 5,500	28	40,001 to 45,000	60
5,501 to 6,000	30	45,001 to 50,000	62
6,001 to 6,500	32	50,001 to 60,000	64
6,501 to 7,000	34	60,001 to 70,000	66
7,001 to 7,500	36	70,001 to 80,000	68
7,501 to 8,000	38	80,001 to 90,000	70
8,001 to 8,500	40	90,001 to 100,000	72
8,501 to 9,000	42	100,001 to 125,000	74
9,001 to 9,500	44	125,001 to 150,000	76
9,501 to 10,000	46	150,001 to 175,000	78
10,001 to 15,000	48	175,001 to 200,000	80
15,001 to 20,000	50	200,001 to 250,000	82
20,001 to 25,000	52	250,001 to 300,000	84 [1/]
25,001 to 30,000	54		

1/ For quantities greater than 300,000 an additional 2 sample batteries shall be selected for each 100,000.

4.6.4.1 Shipment of sample. - Within 24 hours after selection, the batteries shall be stored by the manufacturer until shipment is made. All packages, cartons, and cases containing sample batteries shall be marked:

DO NOT OPEN - TEST SAMPLE

The sample shall be shipped monthly by express, at the manufacturer's expense. The Government will not delay the shipment of any batteries on order pending the results of tests performed outside the manufacturer's plant.

4.6.4.2 Test routine. - Within 2 weeks after the sample is received at the laboratory, the sample will be divided into specified storage and inspection groups. Each group of sample batteries will then be subjected to the tests specified in tables II, III, IV and V, test groups I and III in the order shown.

4.6.4.3 All sample batteries shall be subjected to the visual and dimensional examination in inspection group I. The sample batteries shall then be divided into two equal parts for inspection groups II and III. If

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the number of sample batteries is odd, the extra battery shall be assigned to inspection group III. After the storage period, inspection groups II and III shall each be further divided so that approximately equal quantities of batteries shall be activated and subjected to the specified tests for their particular groups.

4.6.4.4 Noncompliance. - If the API is less than the MPI, calculated as specified in 3.8 and 3.8.1, the shipment release represented by the sample batteries shall be considered as not complying with the requirements of this specification.

4.7 Inspection procedures. -

4.7.1 Visual and dimensional examination. - Each of the sample batteries selected in accordance with 4.6.2 shall be examined and measured to verify conformance to all of the requirements of this specification which do not involve tests.

4.7.2 Activation. - After elapses of the storage period specified (see 3.7), the sample battery shall be activated in accordance with instructions on the label (see 3.5.1.2).

4.7.3 Capacity and voltage tests. - Immediately after activation, in accordance with the instructions on the label (see 3.5.1.2), the sample batteries shall be subjected to tests specified (see 4.8.4).

4.7.4 Drop test. - Immediately following activation and prior to the capacity and voltage tests all the batteries from an inspection group (see 4.7.5) in which the drop test is specified shall be tested in the manner described below. Each battery shall be subjected to 25 arrestments after 6 feet of free fall. The arrestment shall be accomplished with a length of twine. One end of the twine shall be securely attached to a fixed point located at the start of the free fall, and the other end tied by means of a knot to the approximate midpoint of the battery suspension cord. The net length of the arrestment twine shall be 72 ± 1/2 inches.

Table II - Preproduction and acceptance inspection.

Inspection group	Inspection	Storage time at 70° ± 2°F. (21.1° ± 1.1°C.) and 50 ± 5 percent RH	Temperature for discharge through resistance load, after activation
I	Visual and dimensional examination	---	---
II	Drop test	---	---
	Capacity and voltage:		
	Initial low-temperature discharge	30 days	-58° ± 2°F. (-50° ± 1.1°C.)
	Initial high- and low-temperature discharge	30 days	140° ± 2°F. (60° ± 1.1°C.) for 10 minutes, then at -58° ± 2°F. (-50° ± 1.1°C.)
	Initial normal-temperature discharge	30 days	70° ± 10°F. (21.1° ± 5.6°C.)
III	Capacity and voltage:		
	Delayed low-temperature discharge	12 months	-58° ± 2°F. (-50° ± 1.1°C.)
	Delayed high- and low-temperature discharge	12 months	140° ± 2°F. (60° ± 1.1°C.) for 10 minutes, then at -58° ± 2°F. (-50° ± 1.1°C.)
	Delayed normal-temperature discharge	12 months	70° ± 10°F. (21.1° ± 5.6°C.)

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4.7.5 Preproduction and acceptance inspection for individual types.

4.7.5.1 Type BA-253/U. - Preproduction and acceptance inspection for type BA-253/U shall be as specified in table II.

4.7.5.2 Type BA-259/AM. - Preproduction and acceptance inspection for type BA-259/AM shall be as specified in table III.

Table III - Preproduction and acceptance inspection.

Inspection group	Inspection	Storage time at 70° ± 2° F. (21.1 ± 1.1°C.) and 50 ± 5 percent RH	Temperature for discharge through resistance load, after activation
I	Visual and dimensional examination	---	---
II	Capacity and voltage: 1/		
	Initial low-temperature discharge	5 to 30 days	-58° ± 2°F. (-50° ± 1.1°C.)
	Initial high- and low-temperature discharge	5 to 30 days	140° ± 2°F. (60° ± 1.1°C.) for 10 minutes, then at -58° ± 2°F. (-50° ± 1.1°C.)
	Initial normal-temperature discharge	5 to 30 days	80° ± 10°F. (26.7° ± 5.6°C.)
III	Capacity and voltage: 1/		
	Delayed low-temperature discharge	12 months	-58° ± 2°F. (-50° ± 1.1°C.)
	Delayed high- and low-temperature discharge	12 months	140° ± 2°F. (60° ± 1.1°C.) for 10 minutes, then at -58° ± 2°F. (-50° ± 1.1°C.)
	Delayed normal-temperature discharge	12 months	80° ± 10°F. (26.7° ± 5.6°C.)

1/ During these tests, each battery shall be placed within the case used for housing the Radiosonde Modulator MD147( )/AMT-4A, or equal.

4.7.5.3 Type BA-292/AM. - Preproduction and acceptance inspection for type BA-292/AM shall be as specified in table IV.

4.7.5.4 Type BA-316/AM. - Preproduction and acceptance inspection for type BA-316/AM shall be as specified in table V.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Batteries shall be preserved and packaged for the designated level (see 6.1) and in the quantity designated in table VI.

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Table IV - Preproduction and acceptance inspection.

Inspection group	Inspection	Storage time at 70° ± 2°F. (21.1° ± 1.1°C.) and 50 ± 5 percent RH	Temperature discharge through lamp after activation
I	Visual and dimensional examination	---	---
II	Drop test Capacity and voltage: Inital high- and low-temperature discharge	--- 30 days	--- 80° ± 2°F. (26.7° ± 1.1°C.) for 8 minutes, then decrease temperature at a rate of 4°F. (2.2°C.) per minute for 30 minutes. Temperature at the 38th minute to be -40° ± 2°F. (-40° ± 1.1°C.)
III	Drop test Capacity and voltage: Delayed high- and low-temperature discharge	--- 12 months	--- 80° ± 2°F. (26.7° ± 1.1°C.) for 8 minutes, then decrease temperature at a rate of 4°F. (2.2°C.) per minute for 30 minutes. Temperature at the 38th minute to be -40° ± 2°F. (-40° ± 1.1°C.)

5.1.1 Level A. - Batteries shall be individually cushioned by wrapping with 1/2 inch thickness, type II, class B cellulose wadding conforming to Specification PPP-C-843 and preserved and packaged in accordance with method 1A7 of Specification MIL-P-116, using a type III metal can with key attached conforming to Specification PPP-C-96. Plan B exterior coating with olive drab finish shall be required.

5.2 Packing (see 6.1). -

5.2.1 Level A. - Unless otherwise specified (see 6.2), batteries packaged as specified (see 6.2), shall be packed in quantities in accordance with table VI and in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-585	Class 3 use
PPP-B-591	Overseas type
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class 2
PPP-B-576	Class 2

Shipping containers shall have caseliners conforming to Specification MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547. Caseliners for class 2 fiberboard boxes conforming to Specification PPP-B-636 may be omitted provided all corners and edge

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seams and manufacturer's joints are sealed with minimum 2 inch wide tape conforming to Specification PPP-T-76. Boxes shall be closed and banded in accordance with the applicable box specification or appendix thereto. When unit packages are packed in tiers, a fiberboard pad conforming to grade W5 shall be placed between the tiers.

Table V - Preproduction and acceptance inspection.

Inspection group	Inspection	Storage time at 70° ± 2°F. (21.1° ± 1.1°C.) and 50 ± 5 percent RH	Temperature for discharge through resistance load, after activation
I	Visual and dimensional examination	---	---
II	Capacity and voltage:		
	Initial normal-temperature discharge	30 days	75° ± 5°F. (23.9° ± 2.8°C.)
	Initial low-temperature discharge	30 days	-58° ± 2°F. (-50° ± 1.1°C.)
	Initial flight-similitude discharge	30 days	(1/)
III	Capacity and voltage:		
	Delayed normal-temperature discharge	12 months	75° ± 5°F. (23.9° ± 2.8°C.)
	Delayed low-temperature discharge	12 months	-58° ± 2°F. (-50° ± 1.1°C.)
	Delayed flight-similitude discharge	12 months	(1/)

1/ After the battery has been removed from storage, activated in accordance with manufacturer's instructions, and has reached the minimum initial voltage, the battery shall then be inserted in a radiosonde AN/AMT-7 or a substitute box having the same insulating properties as the radiosonde. The radiosonde or the substitute box, with the activated battery mounted therein, shall then be placed in a pressure chamber at ambient room temperature and ambient atmospheric pressure. Simultaneously with the beginning of the discharge, the temperature and pressure shall be reduced at a rate equivalent to an ascension rate of 850 feet per minute in a standard atmosphere. Temperature, pressure, and voltage of each unit shall be read at 10-minute intervals.

Table VI - Unit package and shipping container quantities.

Type	Unit package quantity	Shipping container quantity
BA-253/U	1	72
BA-259/AM	1	24
BA-292/AM	1	50
BA-316/AM	1	12

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5.2.2 Level B. - Unless otherwise specified (see 6.2), batteries packaged as specified (see 6.2), shall be packed in quantities in accordance with table VI and in boxes conforming to any of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-585	Class 1 or 2 use
PPP-B-591	Domestic type
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Class 2
PPP-B-576	Class 1

5.2.3 Level C. - Batteries, preserved and packaged as specified in 5.1, shall be packed in containers in a manner to insure safe delivery and acceptance at destination. Containers shall comply with the Uniform Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking specified in the contract or order or herein, each unit and intermediate package and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type designation (see 1.2).
(c) Number of preproduction samples required and location to which shipment should be made (see 3.1).
(d) Level of preservation, packaging and packing required (see 5.1 and 5.2).

6.2 Adjustment in the event of failure to meet MPI. - In the event API calculated as specified in 3.8.1 is less than 80 percent of MPI (see 3.8) or in the event the weighted average (WA) (see 6.2.3) is 80 percent or more of MPI, and provided the supplier is notified in writing of the results within 6 months after completion of group C inspection, adjustments should be made as specified in 6.2.1.

6.2.1 Adjustments. - Adjustments should be made in cash, or if mutually agreeable between the supplier and the Government, in additional batteries. All such additional batteries should conform to all the requirements of this specification, and should be subject to adjustment as specified herein; however, these adjustments should be made in cash only.

6.2.2 Extent of adjustment on shipment release. - On those shipment releases in which the API is less than 80 percent of the MPI, the adjustment should be determined by the following formula for each such lot:

$$\text{Value of adjustment} = \left(\frac{\text{MPI-API}}{\text{MPI}}\right)$$

Where:

M = The dollar value of the batteries comprising the shipment release. In no case should any shipment release adjustment (whether in cash, additional batteries, or a combination of cash and additional batteries) exceed 70 percent of the contract dollar value of the specific shipment release represented.

6.2.3 Extent of adjustment on WA. - On those shipment releases in which the API is 80 percent or more, but less than the MPI, the adjustment should be determined by the following formula for all such releases:

$$\text{Value of adjustment} = \left(\frac{\text{MPI-WA}}{\text{WA}}\right)$$

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Where:

N = The total-contract dollar value of the batteries minus the total dollar value of the batteries in all the shipment releases which were subject to shipment release adjustment. In the event there are no shipment release adjustments, the N value is the total-contract value of the batteries.

WA = The weighted average performance index computed as follows:

(a) On each shipment lot in which the API is 80 percent or more of the MPI, the number of batteries in each such shipment release is multiplied by its API.
(b) These products are added.
(c) The resulting sum is divided by the total number of batteries on these shipment releases.

No adjustment should result if the computed WA is equal to or greater than the MPI.

6.2.4 In cash. - The total adjustment made in cash for batteries on contract should be equal to the sum of all shipment release adjustments in accordance with 6.2.2, plus the WA adjustment in accordance with 6.2.3.

6.2.4.1 In batteries. - The total adjustment made in batteries should be the quantity of batteries to be furnished as determined by dividing the total adjustment in cash (see 6.2.4.1) by the contract price of each battery.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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Army - SigC
Navy - Ships
Air Force - WADD

Preparing activity:
Navy - Bureau of Ships
(Project 701-191)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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